

Mathematical Modeling (MA206) Spring 2026

Project 1: AI-Assisted Modeling: A Guided Tour

March 5, 2026

Southern University of Science and Technology (SUSTech)

Project Overview

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Project Overview

Embracing AI in Mathematical Modeling

Welcome to Project 1! In this project, we will explore the powerful intersection of generative artificial intelligence and Mathematical Modeling¹.

Core Objective

To understand how Large Language Models (LLMs) can strategically assist in Problem Formulation², Algorithm Design³, and Scientific Writing⁴ without replacing rigorous analytical thinking.

¹Mathematical Modeling: 数学建模

²Problem Formulation: 问题重述

³Algorithm Design: 算法设计

⁴Scientific Writing: 科技论文写作

Task Requirements

Your Task

You are required to select a **2025 CUMCM** (Contemporary Undergraduate Mathematical Contest in Modeling) National First Prize⁵ paper from any country or region.

Your mission is to meticulously reproduce:

- **Core Ideas:** Unpack the underlying Heuristic Algorithms⁶ or statistical assumptions.
- **Mathematical Derivations:** Re-derive the fundamental equations step-by-step.
- **Validation:** Conduct a thorough Sensitivity Analysis⁷ to verify their results.
- **Paper Composition:** Reproduce the layout and presentation of the paper details.

⁵National First Prize: 国家一等奖

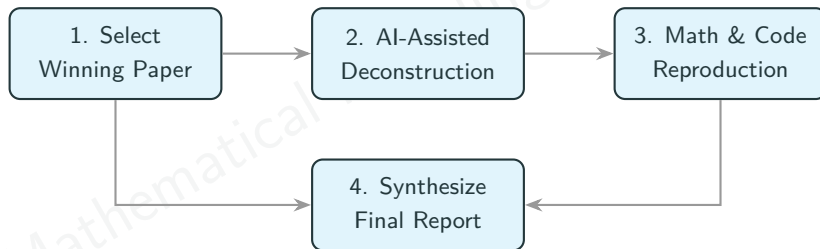
⁶Heuristic Algorithms: 启发式算法

⁷Sensitivity Analysis: 敏感性分析

Recommended Workflow

Project Workflow

Please follow this systematic approach to complete your project:



Deliverables & Evaluation

What you need to submit:

1. A comprehensive Research Report⁸ formatted in LaTeX.
2. Well-documented, reproducible source code (Python, MATLAB, or C++).
3. A brief statement declaring how AI tools were utilized.

Important Note

Your technical usage of AI tools and full papers will be shared among peers.

⁸Research Report: 研究报告

Upcoming Projects and Competitions

Upcoming Competitions and Projects (Projects 2–5)

Project	Host	Approx. Time	Information
MathorCup Challenge	CSOOPem	Mid-April	National Level
“Huazhong Cup” Challenge	Hubei Math Society	Late April	Regional Level
May Day Competition	CUMT / Jiangsu Math Society	Early May	National Level
“Shenzhen Cup” Challenge	CSIAM / Shenzhen Govt.	Jul.–Aug.	High Difficulty

Important Notice

The Shenzhen Cup preliminary round will serve as Project 5, and the other projects will be arranged in the order of their respective competition dates. We encourage everyone to register on time and submit your papers to the competition system.

即将到来的竞赛和项目 (项目 2-5)

赛事名称	主办方	预计时间	相关信息
MathorCup 数学应用挑战赛	中国双法研究会	4 月中旬	国家级竞赛
“华中杯”大学生数学建模挑战赛	湖北省数学学会	4 月下旬	区域性竞赛
五一数学建模竞赛	中国矿大、江苏省工数学会等	5 月上旬	全国性竞赛
“深圳杯”数学建模挑战赛	全国大数模组委会 / 深圳市科协	4 月-8 月	难度高、影响力大

参赛要求与评分

深圳杯初赛将作为 Project 5，其他 project 顺序以比赛时间为准，希望大家及时报名注册并提交论文到赛事系统。

Key Takeaways

- **AI is an Assistant, Not an Oracle:** Use AI to brainstorm and debug, but the mathematical rigor must come from you.
- **Attention to Detail:** National First Prize papers are distinguished by their flawless Robustness⁹ and clear logical flow.
- **Formatting Matters:** Your final submission should look as professional as the papers you are studying.

⁹Robustness: 鲁棒性

Deadline

23:59 on the Sunday of the **third academic week**,
Spring 2026.